Uncompetitive inhibition is rare in one-substrate reactions but common in two-substrate reactions.

The degree of uncompetitive inhibition may increase with increase in substrate concentration.

The inhibitor constant (K_i) of uncompetitive inhibition is

$$K_i = \frac{([ES]\ [I])}{([ESI])}$$

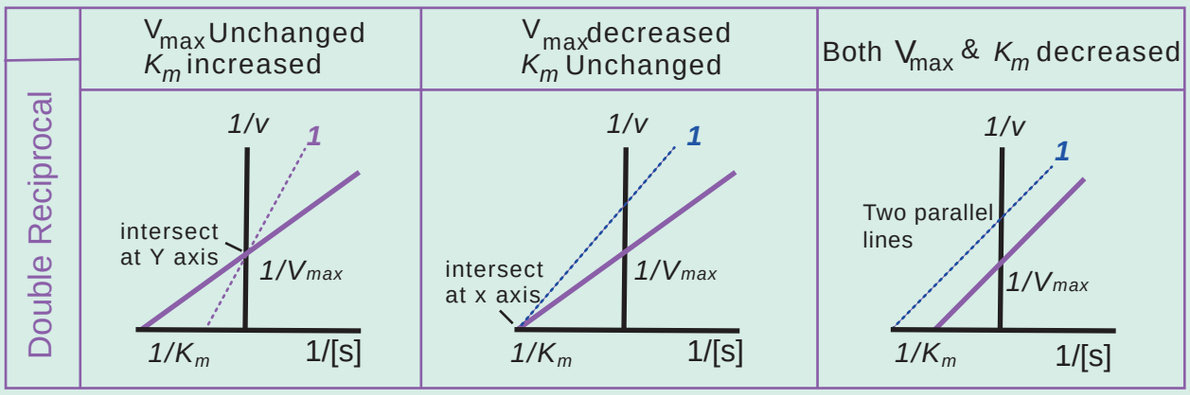


Figure 9.12 Lineweaver Burk plots for competitive, non-competitive and uncompetitive inhibition.

Differences Between Competitive And Noncompetitive Inhibition

COMPETITIVE INHIBITOR	NON-COMPETITIVE INHIBITOR
Inhibitor resembles the substrate (substrate analogue)	Inhibitor has no structural resemblance with the substrate.
Inhibitor binds at the active site.	Inhibitor binds at a site other than active site
Enzyme binds either with substrate or inhibitor	Enzyme binds with both substrate and inhibitor
Reversible	Irreversible
Can be overcome by increasing substrate concentration.	Cannot be overcome by increasing substrate concentration
K_m value will be increased	K_m remains unchanged
V_{max} remains unchanged.	V_{max} value will be decreased

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